A. S. M. SAMIUL ISLAM

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LinkedIn | GitHub

Research Interests

HCI and Accessibility • User Experience Research • Embedded Systems • TinyML • Machine Learning • WiFi Sensing • Digital Health • Privacy-Preserving Machine Learning

Education

Master of Science in Management of Technology (MOT) October 2024 – Present
Institute of Appropriate Technology (IAT),
Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh
• CGPA: 3.67/4.00 | Thesis defence pending
• Thesis: IoT and LLM-Based Bangla Sign Language Recognition Using a Wearable Smart Glove (proposal approved by BUET's Committee for Advanced Studies and Research)
• Supervisor: Tahsina Farah Sanam, Ph.D.

Bachelor of Science in Computer Science and Engineering 2019 – 2023
North South University, Dhaka, Bangladesh
• CGPA: 3.59/4.00 | 50% Merit Scholarship | Distinction: **CUM LAUDE**

Higher Secondary Certificate (HSC), Science July 2016 – June 2018
Govt. Azizul Haque College, Bogura, Bangladesh | GPA: 5.00/5.00

Secondary School Certificate (SSC), Science February 2011 – June 2016
Bogra Zilla School, Bogura, Bangladesh | GPA: 5.00/5.00

Professional Experience

Research Assistant (Part-time), ScabieSense Project January 2026 – Present
Cyber Physical Intelligence (CPI) Lab, IAT, BUET; appointed through the Research and Innovation Centre for Science and Engineering (RISE), BUET
• Project: "From Rash to Remedy: Tackling Scabies with a One-Stop Digital Healthcare Solution"; PI: Tahsina Farah Sanam, Ph.D.
• Designed the clinical-AI pipeline (questionnaire scoring, image analysis, fallback logic) and the mobile UX; the tool is now used by doctors screening madrasa students

Postgraduate Research Fellow November 2024 – Present
Cyber Physical Intelligence (CPI) Lab, Institute of Appropriate Technology, BUET, Dhaka, Bangladesh
• M.Sc. thesis: a sensor glove that recognizes Bangla Sign Language, with an offline language model that builds sentences
• WiFi CSI sensing for early autism screening, including a paper accepted at IEEE TENCON 2026

UI/UX Designer and Researcher August 2023 – April 2024
SongEun Research and Development (SRRD), Mirpur DOHS, Dhaka (Korea-operated)
• Designed AI-powered CCTV-based Video Software as a Service (VSaaS) platform
• Created user flows, schema diagrams, and game UI integrated with VSaaS system
• Designed user diagrams for connected online chat software

Teaching Assistant, Department of Mathematics and Physics June 2021 – January 2022
North South University, Dhaka, Bangladesh
• Assisted five professors in course delivery and student assessment; advised undergraduate students

Achievements

• **Creative Visionary** (Certificate of Achievement), Shahid Smrity Hall, BUET, for end-to-end design contributions to the Intra Hall Games 2026 (August 2026)
• **Best Poster Design Award**, Inter-Hall July Uprising Competition, BUET
• **Startup Incubator Selection**, ICSETEP Project (Asian Development Bank-funded), Department of CSE, BUET (August 2025)
• **UIHP Pre-seed Grant**, Team Arora, Cohort IC5, University Innovation Hub Program at BUET

(2024)
• **CUM LAUDE Distinction**, North South University (2022)
• **50% Merit Scholarship**, North South University (2019–2022)
• **17th Position**, Regional 3rd Bangladesh Physics Olympiad
• **Divisional 2nd Runner Up**, 9th and 10th Bangladesh Math Olympiad
• **First Position (Team)**, Acting Competition, BTV Subarna Jayanti

Research Experience

IoT and LLM-Based Bangla Sign Language Recognition with a Smart Glove Ongoing
M.Sc. Thesis, CPI Lab, IAT, BUET (Supervisor: Tahsina Farah Sanam, Ph.D.)
• Built a sensor glove with four flex sensors and an MPU-6050 motion sensor; recorded 1,369 samples of 23 Bangla Sign Language signs from one signer, 12 of them mental-health terms
• A 1D CNN recognized signs with 97.2% accuracy in 0.37 ms per sign; on unseen recording sessions macro-F1 fell to 0.733, and the motion sensors alone generalized better than the full glove
• An offline language model turns recognized signs into a sentence, spoken only after the user confirms; first-author manuscript almost ready to submit
• Next version (in progress): a lighter glove with six motion sensors and no flex sensors, built on comfortable gaming gloves that fit any hand size and redesigned after user-experience research; HCI studies under way

SMMNet-FedHE: Privacy-Preserving Framework for Stereotypical Motor Movement Monitoring with WiFi CSI Accepted, IEEE TENCON 2026
CPI Lab, IAT, BUET, with A. Islam and T. F. Sanam
• Combined federated learning and homomorphic encryption for privacy-preserving, contactless detection of stereotypical motor movements for early autism screening
• Extracted multi-domain features from WiFi CSI data; accuracy above 80% across centralized and federated settings

Wi-SMM: Stereotypical Motor Movement Detection for Autism Screening Using WiFi CSI Submitted
CPI Lab, IAT, BUET, with M. Mohtasim, A. Islam, T. F. Sanam
• Built a WiFi CSI-based dataset and detection framework for early, contactless ASD screening
• New portable setup that collects CSI data remotely and moves easily between sites

Wi-Fi CSI-Based Human Activity Detection Ongoing
Independent Project
• Capturing Wi-Fi channel state information (CSI) with two ESP32 boards for activity classification, with firmware built in ESP-IDF

ScabieSense: Scabies Detection, Treatment Guidance, and Outbreak Mapping Ongoing
CPI Lab, IAT, BUET, with the Bangladesh Academy of Dermatology (BAD); RISE-BUET project supported by the International League of Dermatology and BAD
• Researched the screening, case tracking, and follow-up workflow; designed the clinical-AI pipeline (questionnaire scoring, image analysis, fallback logic)
• Focused on mobile-friendly user experience and secure role-based access

BD Pedestrian and Overtaking Action Recognition for Driver Assistance
Undergraduate Research Project, North South University
• Collected and annotated 10+ hours of video dataset for deep learning models
• Developed computer vision algorithms for real-time action recognition for driver assistance in Bangladesh

Hakuna Matata – Mental Health Social Network Platform
Undergraduate junior Research Project, North South University
• Designed user-centered interfaces using HTML, CSS, and JavaScript
• Conducted user research on community engagement and applied HCI principles for accessibility

Publications and Manuscripts

Conference Proceeding
• A. Islam, **A. S. M. S. Islam**, T. F. Sanam, "SMMNet-FedHE: Privacy-Preserving Deep Learning-Based Framework for Stereotypical Motor Movement Monitoring with WiFi CSI Data," *IEEE TENCON 2026* [Accepted].

Manuscripts

• **A. S. M. S. Islam**, T. F. Sanam, et al., "Real-Time Sensor-Glove Recognition and Language-Model Sentence Building for Bangla Sign Language Mental-Health Vocabulary, and What the Accuracy Measures" [Manuscript almost ready to submit].

• M. Mohtasim, A. Islam, **A. S. M. S. Islam**, T. F. Sanam, "Wi-SMM: Detection of Stereotypical Motor Movements for Early Screening of Autism Spectrum Disorder Using a WiFi CSI-Based Dataset" [Submitted].

• **A. S. M. S. Islam** (third author; author list anonymized for review), "OpenEyeGaze: In-Browser Gaze Estimation with Planar Head-Motion Normalization, Online Adaptation, and Zero-DOM Reading Telemetry" [Submitted to ETRA 2027: ACM Symposium on Eye Tracking Research & Applications].

Service

University Service

Vice President, NSU HR Club, North South University — March 2022 – October 2023

• Organized "HR SPEAKS OUT 4.0" seminar with 700+ attendees

• Led IT, promotional, and strategic planning activities

• Managed recruitment of 300+ members

Sub-Executive (November 2020 – March 2022) | *Senior Member* (March 2020 – November 2020)

Community Service

Panel Deputy Speaker, Child Parliament of NCTF, Bangladesh — May 2016 – June 2018

• Prepared speakers and facilitated pitches to the Education Minister and Deputy Speaker of the National Parliament

Trainer and UDRT Member, Bangladesh Red Crescent Society, Bogra Unit — March 2015 – August 2016

• Completed Canadian Red Cross disaster management training; led relief operations in Sariakandi, Bogura

Unit Leader, Bogra Zilla School Red Crescent Unit — October 2013 – October 2015

• Led team of 100+ active and 300 regular members

• Coordinated relief aid to 100+ flood-affected families

Languages and Skills

Languages: Bengali (native), English (professional proficiency)
Programming: Python, C/C++, Java, JavaScript, Swift
Web Technologies: HTML, CSS, Bootstrap, SQL
Mobile Development: iOS (Swift, Xcode)
Design: UI/UX Design, Figma, Wireframing, Prototyping, User Research, AutoCAD (2D/3D)
Embedded Systems: ESP32 (ESP-IDF), Arduino Uno, NodeMCU, MPU-6050, TCA9548A, Wi-Fi CSI
ML and Data: TensorFlow, TenSEAL (homomorphic encryption), NumPy, Pandas, Google Colab
Tools: Git, MS Office Suite, Google Workspace, LaTeX

Professional Development

• **Supervised Machine Learning: Regression and Classification**, DeepLearning.AI & Stanford University, Coursera — August 2025

• **Google UX Design Professional Certificate**, Google, Coursera (UX foundations, wireframing, prototyping, Figma)

• **The Complete 2021 Web Development Bootcamp**, Udemy — September 2021

References

Tahsina Farah Sanam, Ph.D.
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Nabeel Mohammed, Ph.D.
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